Tutorial 1

Maps, Vector spaces and Duality.

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1. Decompose the given map f as a composition of elementary operations, such as squaring. Define each elementary map using the $A \to B;\ a \mapsto b$ notation.

$$f(x) = \sqrt[3]{\cos(x^2)}, \qquad x \in \mathbb{R}$$

For each elementary map, specify whether it is injective, surjective, bijective or none with the domain A and target (codomain) B you defined for the map.

2. Tick the true statements:

◯ Every vector is a matrix with only one column.

◯ Every linear map between vector spaces can be represented by a unique quadratic matrix.

◯ Every vector space has a unique corresponding dual vector space.

◯ Every vector in a vector space has a unique dual covector in the dual space.

◯ One can always reconstruct a vector from its components and the corresponding basis.

◯ A change of basis in a vector space does not change a vector in that space.

◯ Linearity of a map $\varphi\colon V \to W$ depends on the choice of basis in V and W.

◯ Given a basis of a vector space, one can compute components of covectors in the dual space with respect to the dual basis.

◯ The length of a vector $(3, 4, 5) \in \mathbb{R}^3$ with the standard vector space structure is $\|v\| = \sqrt{50}$.

3. Notation reminder:

Space	V	V^*
Elements	$v \in V$	$\alpha \in V^*$
Basis elements	e_i	ε^j
Components	$v^i \in \mathbb{R}$	$\alpha_j \in \mathbb{R}$
Decomposition	$v = v^i e_i$	$\alpha = \alpha_j \varepsilon^j$
Matrix notation	$\boldsymbol{v} = \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix}$	$\boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 & \cdots & \alpha_n \end{pmatrix}$

⚠ Pay attention: some people arrange covector components not in a row but in a column. This way, they get expressions in components like $P = \boldsymbol{\alpha}^\top \cdot \boldsymbol{v}$, where $\boldsymbol{\alpha}^\top$ is now a row. If the components are arranged as a row (as in the table), we have $P = \boldsymbol{\alpha} \cdot \boldsymbol{v}$. The transformation of components of a covector is written with the row notation (as in the table) as:

$$\tilde{\boldsymbol{\alpha}} = \boldsymbol{\alpha} \cdot A \qquad \text{or} \qquad \tilde{\boldsymbol{\alpha}}^\top = A^\top \cdot \boldsymbol{\alpha}^\top$$

(the second equation is the transpose of the first one) instead of

$$\tilde{\boldsymbol{\alpha}} = A^\top \cdot \boldsymbol{\alpha}$$

with the column notation. Look at the lecture notes for a detailed explanation/derivation and compare with the slides.

4. Consider the composition $\varphi \circ \theta$ of two linear maps $\theta \colon U \to V$ and $\varphi \colon V \to W$ with U, V and W having the dimensions k, n and m, respectively. Let $(g_1, \ldots, g_l)$, $(e_1, \ldots, e_n)$ and $(f_1, \ldots, f_m)$ be bases in U, V and W. Show that the coordinate representation of the composition $\varphi \circ \theta$ is given by the matrix product $\Phi \cdot \Theta$ of the coordinate representations Φ of φ and Θ of θ. First, obtain an expression in components and then verify that the matrix multiplication rule 'row times column' replicates this expression.

5. Let $(V, \oplus, \odot)$ be a vector space and let $\varphi \colon V \to V^*$ and $\psi \colon V \to V$ be linear maps. Prove that the map

$$\begin{aligned} \pi \colon V \times V &\to \mathbb{R} \\ (v, w) &\mapsto \varphi(v)\big(\psi(w)\big) \end{aligned}$$

is a bilinear map on V. (Bilinear means independently linear in each slot.)

6. Consider the vector space $(\mathbb{R}^2, +, \cdot)$ with the standard addition and scalar multiplication and two vectors $e_1 = (1, 2)$ and $e_2 = (3, 4)$.

   (a) Is the map

   $$\begin{aligned} f \colon \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto 3x + 2y \end{aligned}$$

   a linear map? Of what space f is an element?

   (b) Check that the vectors e_1 and e_2 form a basis in $\mathbb{R}^2$.

   (c) Find the basis $(\varepsilon^1, \varepsilon^2)$ of $(\mathbb{R}^2)^*$ dual to the basis (e_1, e_2).

   (d) Find the components of f in the dual basis, i.e. the numbers $f_1, f_2 \in \mathbb{R}$ such that $f = f_1 \cdot \varepsilon^1 + f_2 \cdot \varepsilon^2$.

7. Consider the vector space $(V, +, \cdot)$, where V is the space of second order polynomials on the interval $[-1, 1]$, i.e. any element of V is of the form

$$\begin{aligned} f \colon [-1, 1] &\to \mathbb{R} \\ x &\mapsto ax^2 + bx + c \end{aligned}$$

where $a, b, c \in \mathbb{R}$, and the operations are defined point-wise. Consider as basis vectors $e_1 = (x \mapsto x^2)$, $e_2 = (x \mapsto x)$ and $e_3 = (x \mapsto 1)$. The components of any vector $f(x) = ax^2 + bx + c$ with respect to the chosen basis are $f^1 = a$, $f^2 = b$ and $f^3 = c$.

(a) Why is this vector space 3-dimensional?

(b) Compute the dual basis $(\varepsilon^1, \varepsilon^2, \varepsilon^3)$ with respect to the chosen basis.

(c) We define the *Dirac delta* $\bar{\delta}$ as the operator acting on an element of V and extracting its value at $x = 0$, i.e. $\bar{\delta}(f) := f(0)$. Show that $\bar{\delta}$ is a co-vector, i.e., $\bar{\delta} \in V^*$.

(d) What is the component representation of $\bar{\delta}$ with respect to the previously computed dual basis?